

Software Requirements Specification and Design.

Eagle Eye logistics: *An Automated Web-Based Courier & Consignment Tracking System.*

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Team Eagle Eye	30 Jan 2026	Initial SRS creation	1.0
Team Eagle Eye	27 Feb 2026	Added Design, Security, and Traceability sections	1.1



Table of Contents

1.0 Introduction

- 1.1 Purpose
- 1.2 Scope
- 1.3 Glossary
- 1.4 References
- 1.5 Overview

2.0 Overall Description

- 2.1 System Environment
- 2.2 Functional Requirements Specification
- 2.3 User Characteristics and classes
- 2.4 Non-Functional Requirements

3.0 Requirements Specification

- 3.1 External Interface Requirements
- 3.2 Functional Requirements
- 3.3 Non-functional Requirements
- 3.4 Software Quality Attributes
- 3.5 Other Requirements

4.0 Software Design

- 4.1 System Architecture
- 4.2 Detailed Design
 - 4.2.1 Class Diagrams
 - 4.2.2 Sequence Diagrams
- 4.3 Database Schema
- 4.4 Security Design

5.0 Traceability

- 5.1 Requirement-to-Design Mapping

1.0. Introduction

1.1. Purpose

The purpose of this document is to present a detailed description of the **Eagle Eye Web-Based Tracking System**. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate, and how the system will react to external stimuli. This document is intended for both the stakeholders and the developers of the system.

1.2. Scope of Project

This software system will be a Web-based Logistics Management System. The system is designed to maximize the logistics hub's efficiency by providing tools to assist in automating the consignment booking, tracking, and delivery process.

More specifically, this system is designed to allow an Admin to manage hubs and routes, while allowing Customers to track parcels via an interactive map. The software will facilitate communication between the System and the Customer via Email/SMS alerts.

1.3. Glossary

Term	Definition
AWB	Air Waybill; A unique alphanumeric code used to track specific consignment.
Consignment	The physical package or parcel being transported.
Hub	A central warehouse or sorting facility where parcels are processed.
Last-Mile	The final leg of the delivery process is from the destination hub to the customer.
POD	Proof of Delivery (Digital Signature).
User	Refers to either the Customer, Admin, or Delivery Agent.

1.4. References

- IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications.
- OpenStreetMap / Leaflet JS API Documentation.

1.5. Overview of Document

The next chapter, the Overall Description section, of this document gives an overview of the functionality of the product. It describes the informal requirements and is used to establish a context for the technical requirements specification in the next chapter.

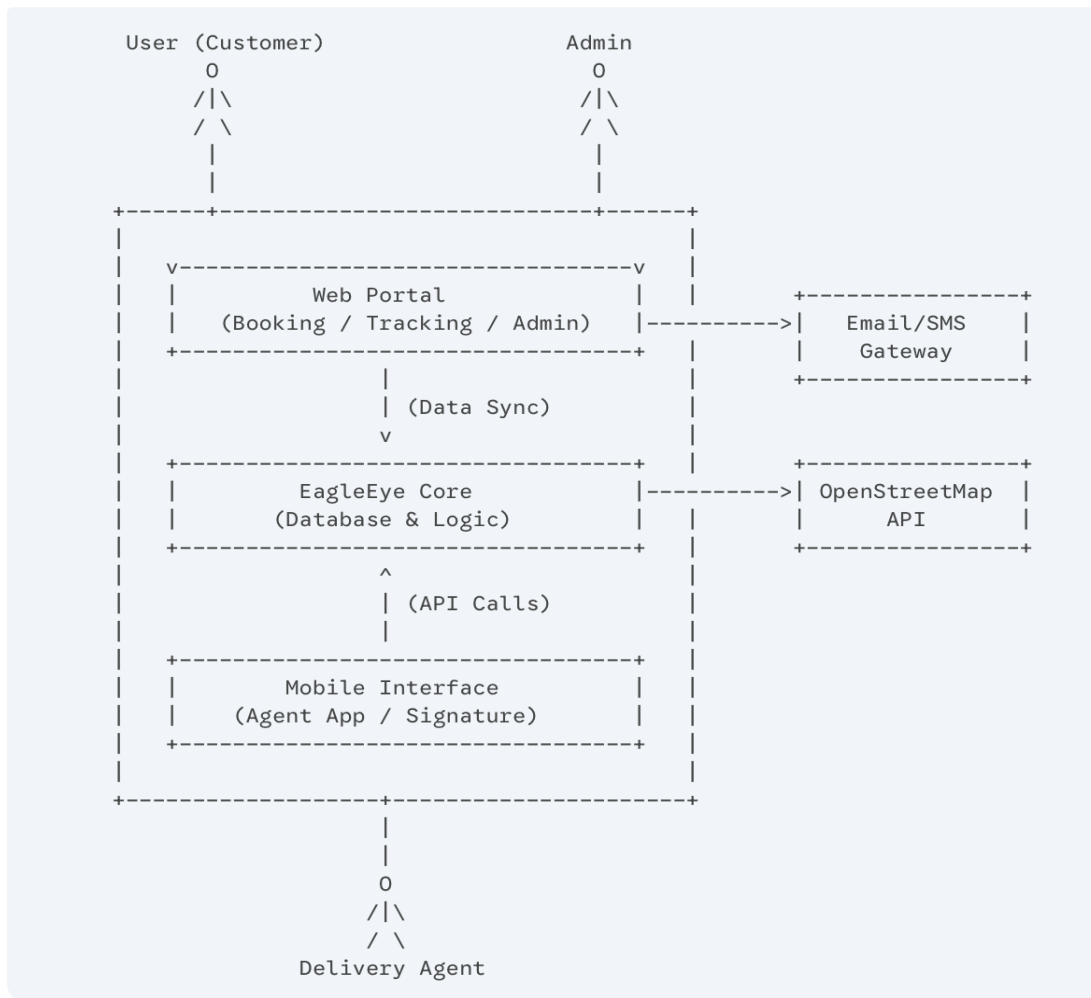
The third chapter, the Requirements Specification section, of this document is written primarily for the developers and describes in technical terms the details of the functionality of the product.

Both sections of the document describe the same software product in its entirety but are intended for different audiences and thus use different languages.

2.0. Overall Description

2.1 System Environment

Figure 1 - System Environment



The Eagle Eye System has three active actors and two cooperating external systems. The Customer and Administrator access the **Web Portal** through the Internet via a web browser. The Delivery Agent accesses the system's functionality through the **Mobile Interface** to update status and capture signatures.

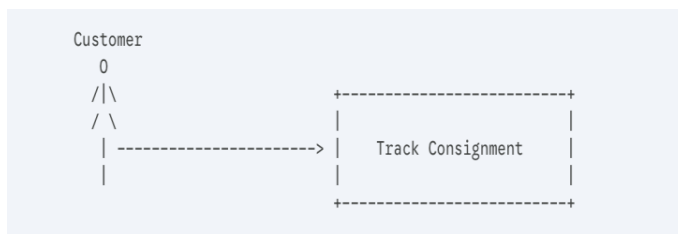
The system interacts with the **OpenStreetMap API** to render tracking maps. Automated communications are routed through an external **Email/SMS Gateway**. There is a link to the internal Database for storing consignment records.

2.2 Functional Requirements Specification

This section outlines the use cases for each of the active actors separately. The Customer and Delivery Agent have specific focused use cases, while the Administrator manages the configuration of the system.

2.2.1 Customer Use Case

Use case: Track Consignment Diagram:



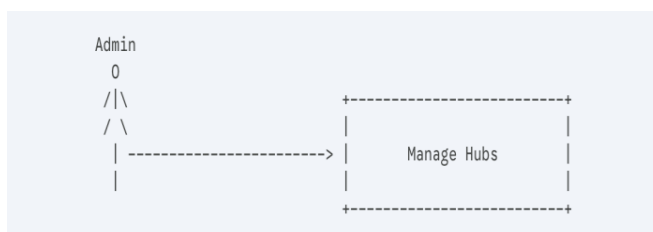
Brief Description the Customer accesses the Eagle Eye Web Portal, searches for a consignment using the AWB number, and views the live location on the map.

Initial Step-By-Step Description Before this use case can be initiated, the Customer has already accessed the Eagle Eye Website.

1. The Customer chooses to track a consignment.
2. The system displays a search bar for the Tracking ID (AWB).
3. The Customer enters the AWB and submits.
4. The system validates the ID and retrieves the route history.
5. The system displays the interactive map with the current package location.
6. The Customer clicks on markers to view timestamp details.

2.2.2 Admin Use Case

Use case: Manage Hubs Diagram:



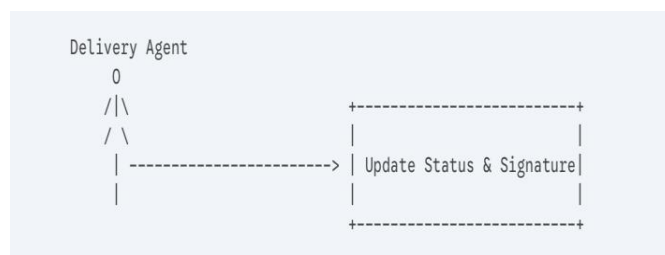
Brief Description the Administrator adds or updates logistics hubs and assigns Pin code mappings to them.

Initial Step-By-Step Description Before this use case can be initiated, the admin has already logged into the Admin Dashboard.

1. The admin selects the "Hub Management" module.
2. The system displays a list of existing hubs.
3. The admin chooses "Add New Hub".
4. The admin enters the Hub Name, Address, and associated Pin code.
5. The system validates the Pin code and saves the Hub to the database.
6. The system returns the admin to the main dashboard.

2.2.3 Delivery Agent Use Case

Use case: Update Status & Signature Diagram:



Brief Description the Delivery Agent updates the status of a parcel to "Delivered" and captures the customer's digital signature.

Initial Step-By-Step Description Before this use case can be initiated, the Agent is logged into the Mobile Interface and is at the delivery location.

1. The Agent selects the parcel from their "Run Sheet".
2. The Agent taps "Mark Delivered".
3. The system presents a digital signature canvas.
4. The Customer signs on the screen.
5. The Agent submits the form.
6. The system updates the status to "Delivered" and triggers the completion of the alert.

2.2.4 Admin Management Use Cases

> Use case: Register Delivery Agent Brief Description:



Brief Description the Admin creates a new account for a Delivery Agent to allow them access to the Mobile Interface.

Initial Step-By-Step Description Before this use case can be initiated, the admin has already logged into the Web Portal.

1. The admin selects "Manage Agents" from the dashboard.
2. The admin chooses "Add New Agent".
3. The admin enters the Agent's Name, Phone Number, and Assigned Hub.
4. The system saves the Agent's details in the Database.
5. The system automatically sends a login password via the external SMS Gateway to the Agent.

> Use case: Monitor Network Status:



Brief Description the Admin views the live dashboard to check pending deliveries, active hubs, and daily revenue.

Initial Step-By-Step Description Before this use case can be initiated, the admin has already logged into the Web Portal.

1. The admin selects "Live Status" from the main menu.
2. The system queries the Eagle Eye Core Database for all active packages.
3. The system returns a list of parcels categorized by "In Transit", "Delivered", and "Delayed".
4. The system calculates and displays the total revenue for the day.
5. The system returns the admin to the main dashboard.

> Use case: Send Broadcast Alert Brief Description:



Brief Description the Admin sends a bulk notification (e.g., "Heavy Rain Delay") to a specific group of recipients, such as all Delivery Agents or all Customers in a specific region.

Initial Step-By-Step Description Before this use case can be initiated, the admin has already logged into the Web Portal.

1. The admin selects the "Notification Center" from the dashboard.
2. The admin selects the recipient group (e.g., "All Agents" or "Delhi Customers").
3. The admin types the alert message (e.g., "Service delayed due to weather").
4. The admin clicks "Send."
5. The system pushes the message to the external **Email/SMS Gateway**.
6. The system confirms the delivery count and returns the admin to the dashboard.

2.3 User Characteristics

The **Customer** is expected to be Internet literate and comfortable using standard web browsers (such as Chrome, Firefox, or Safari) on both desktop and mobile platforms. They must be capable of entering alphanumeric tracking codes into search fields and interpreting visual data presented on interactive maps (zooming, panning). No specific technical training is required, but familiarity with standard e-commerce workflows, such as filling out booking forms and using digital payment gateways, is assumed.

The **Administrator** is expected to be proficient in using desktop computing environments and have a strong command of web-based administrative dashboards. They must be capable of navigating complex data tables, utilizing drop-down menus, and generating analytical reports. While they do not need programming skills, a functional understanding of logistics terminology (e.g., Hubs, Pin codes, Routes) is necessary to effectively configure the system parameters and manage the delivery network.

The **Delivery Agent** is expected to be proficient with smartphone technology, specifically Android or iOS devices. They must be comfortable using touch-screen interfaces to navigate the mobile application. Essential skills include the ability to use GPS navigation tools to locate addresses, operate the device's camera for scanning barcodes, and manage mobile data connectivity. They must also be capable of handling digital signature capture ("Sign-on-Glass") interactions with customers at the point of delivery.

2.4 Non-Functional Requirements

The **Eagle Eye Web Portal** will reside on a cloud-based server with high-speed Internet capability to support real-time map rendering and concurrent user access. The physical machine to be used will be determined by the hosting provider (e.g., AWS/Heroku). The software developed here assumes the use of a backend framework such as **Node.js or Python for** the connection between the Web pages and the database.

The Eagle Eye Database will be a relational database (PostgreSQL via Prisma ORM) capable of handling transactional records for consignments and users.

The Delivery Agent interface is a responsive web view accessible on any modern browser and requires internet connectivity. The speed of the Agent's connection will depend on the mobile network coverage; however, the application is designed to queue status updates if the connection is temporarily lost.

3.0. Requirements Specification

3.1 External Interface Requirements

3.1.1 Software Interfaces The primary link to an external system is the link to the **OpenStreetMap API** (or Google Maps API) to render the visual tracking path. The **Track Consignment** use case sends the latitude and longitude coordinates of the visited hubs to the API, which returns the map tiles and polyline overlay for display to the Customer.

The system also interfaces with an **Email/SMS Gateway** (such as Twilio or a standard SMTP Server) to send automated notifications. The **Update Status** and **Send Broadcast Alert** use cases send the recipient's contact details and the message body to the Gateway, which returns a delivery confirmation status.

3.1.2 Hardware Interfaces the Mobile Interface for the Delivery Agent requires direct interaction with the device hardware:

1. **Camera:** The **Update Status** use case accesses the device camera to scan the AWB Barcode. The system interprets the visual barcode as a unique alphanumeric string (AWB ID).
2. **Touchscreen (Digitizer):** The **Proof of Delivery** use case utilizes the touchscreen to capture the customer's signature. The system records the touch coordinates as an image file.
3. **GPS Receiver:** The mobile application accesses the device's Global Positioning System to fetch the current latitude and longitude when an Agent updates a parcel status.

3.2 Functional Requirements

3.2.1 Book Consignment

Use Case Name	Book Consignment
Trigger	The Customer selects "Book Parcel" on the main website.
Precondition	The Customer is logged in or on the booking page.
Basic Path	1. The Customer enters sender & receiver details (name, phone, pincode). 2. System Logic: The system queries the Pincode API to identify the "Source Hub" linked to the Pin code.

	3. System Logic: The system calculates the distance using the coordinates and applies to the Pricing Algorithm. 4. The System displays the Estimated Cost. 5. The Customer confirms and pays. 6. The System generates a unique AWB (Tracking ID) and displays the printable label.
Alternative Paths	In Step 2, if the Pin code is not mapped to any Hub, the system displays "Service Not Available" and aborts the booking.
Postcondition	A new Consignment record is created in the database with the status "Booked".
Exception Paths	The Customer may abandon the booking at any time.

3.2.2 Track Consignment

Use Case Name	Track Consignment
Trigger	The Customer enters a Tracking ID (AWB) on the homepage.
Precondition	The Customer has a valid AWB number.
Basic Path	1. The Customer submits the AWB. 2. The System queries the Tracking History table. 3. The System retrieves latitude/longitude for all visited Hubs. 4. The System calls the OpenStreetMap API to render the Map and draws a polyline connecting the points. 5. The System displays the current status text (e.g., "Out for Delivery") and the timeline of events. 6. The Customer clicks markers to see specific timestamps.

Alternative Paths	None.
Postcondition	The Map and Status Timeline are displayed.
Exception Paths	If the AWB is invalid, the system shows "Record Not Found."

3.2.3 Update Status

Use Case Name	Update Status
Trigger	The Agent enters AWB number at their Hub
Precondition	The Agent is logged into the Mobile View.
Basic Path	1. The Agent selects "Scan Parcel" and enters the AWB number manually 2. The System validates the AWB ID and searches DB 3. The System verifies the ID exists in the database. 4. The Agent selects the new status (e.g., "Arrived at Hub" or "Out for Delivery"). 5. The System updates the database with the current Timestamp, Hub Location, and Agent ID. 6. The System triggers an email notification to the Customer.
Alternative Paths	Manual Entry: In Step 1, if the barcode is damaged, the Agent manually types the AWB ID via the keypad.
Postcondition	The parcel status is updated in the database.
Exception Paths	If the network is unavailable, the system queues the update and syncs when connectivity is restored.

3.2.4 Capture Proof of Delivery

Use Case Name	Capture Proof of Delivery
Trigger	The Agent is at the customer's doorstep delivering the package.
Precondition	The Status is set to "Out for Delivery".
Basic Path	1. The Agent selects "Mark Delivered" for the specific AWB. 2. The System opens a digital signature canvas on the touchscreen. 3. The Customer signs on the screen. 4. The Agent clicks "Submit". 5. The System saves the signature as an image file linked to the Order ID. 6. The System marks the order as "Completed".
Postcondition	The Order is closed, and the signature is saved in the database.
Exception Paths	The Agent may cancel the delivery attempt if the customer is not available (Status becomes "Undelivered").

3.2.5 Register Delivery Agent

Use Case Name	Register Delivery Agent
Trigger	The admin selects to add a new Agent to the workforce.
Precondition	The admin is logged into the Web Portal.
Basic Path	1. The admin selects "Add New Agent". 2. The admin enters the Name, Phone Number, and assigns a specific Hub. 3. The System validates the phone number format. 4. The System creates a new Agent Record in the User Database.

	5. The System sends a login OTP via SMS for the Agent to verify to the Agent.
Alternative Paths	In Step 3, if the phone number already exists, the system displays "User Already Registered."
Postcondition	The Agent can now log in to the Mobile Interface.

3.2.6 Monitor Network Status

Use Case Name	Monitor Network Status
Trigger	The admin navigates to the "Live Status" dashboard.
Precondition	The admin is logged into the Web Portal.
Basic Path	1. The System queries the database for all active consignments. 2. The System groups the data by status (e.g., "In Transit", "Delayed"). 3. The System calculates total daily revenue. 4. The System renders graphs and charts on the dashboard. 5. The admin views the live data.
Postcondition	The admin is presented with the current operational metrics.

3.2.7 Send Broadcast Alert

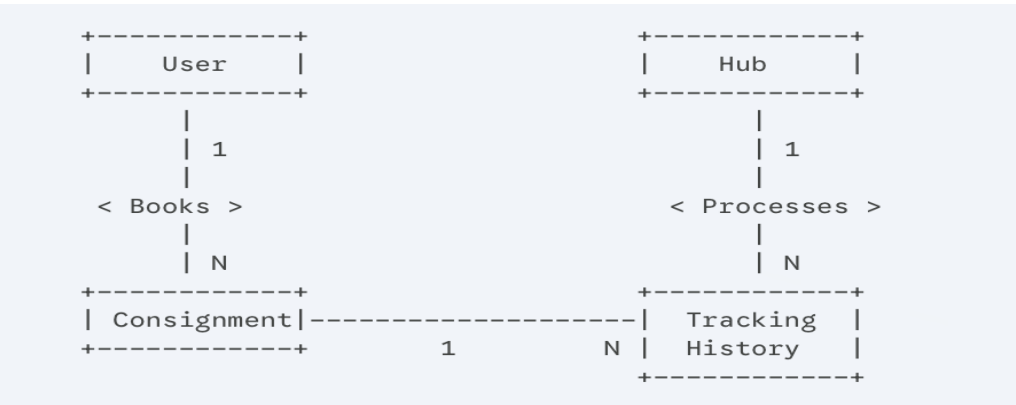
Use Case Name	Send Broadcast Alert
Trigger	The admin needs to notify a group of users about an event (e.g., weather delay).
Precondition	The admin is logged into the Web Portal.
Basic Path	1. The admin selects the target group (e.g., "All Agents" or "Customers in Delhi"). 2. The admin types the alert message. 3. The admin clicks "Send".

	4. The System retrieves contact details for all users in the target group. 5. The System iterates through the list and pushes the message to the SMS/Email Gateway. 6. The System displays a success count (e.g., "Sent to 50 users").
Postcondition	Notifications are queued for delivery.

3.3 Detailed Non-Functional Requirements

3.3.1 Logical Structure of the Data

The logical structure of the data to be stored in the internal **Eagle Eye Core Database** is given below.



The data descriptions of each of these data entities is as follows:

> Consignment Data Entity

Data Item	Type	Description	Comment
AWB	String	Unique Tracking ID (Primary Key)	Alphanumeric (e.g., EE12345)
Sender_ID	Integer	Link to Customer (User) Entity	Foreign Key
Weight	Float	Weight in Kilograms	Used for cost calculation
Current_Status	Text	Current state of the parcel	e.g., "In Transit", "Delivered"

Source_Hub	Integer	Link to Hub Entity	Origin Point
Dest_Hub	Integer	Link to Hub Entity	Destination Point
Order_Date	DateTime	Timestamp of booking	System generated
Signature_Img	Blob	Digital Signature Image	Null until delivery is complete

> Hub Data Entity

Data Item	Type	Description	Comment
Hub_ID	Integer	Unique Hub ID (Primary Key)	Auto-incremented
Hub_Name	Text	Name of the facility	e.g., "Delhi Central Hub"
Address	Text	Physical Address	
Pincode	Integer	The Pincode this Hub services	Used for routing logic
Latitude	Float	Geo-coordinate	For Map plotting via API
Longitude	Float	Geo-coordinate	For Map plotting via API

> User Data Entity (Agents & Customers)

Data Item	Type	Description	Comment
User_ID	Integer	Unique User ID (Primary Key)	
Role	Text	Type of user	"Admin", "Agent", or "Customer"
Name	Text	Full Name	
Phone	String	Contact Number	Used for SMS Alerts

Email	String	Email Address	Used for Login
Password_Hash	String	Encrypted Password	Security Requirement (SHA-256)

> Tracking History Entity

Data Item	Type	Description	Comment
Track_ID	Integer	Unique Record ID	
AWB	String	Link to Consignment	Foreign Key
Hub_ID	Integer	Link to Hub Location	Where the scan happened
Scan_Time	DateTime	Time of scan	
Activity	Text	Description of event	e.g., "Departed Facility"

3.3.2 Security

The server on which the **Eagle Eye Web Portal** resides will have its own security to prevent unauthorized write/delete access. Access to the administrative dashboard is strictly restricted to authorized personnel (Admins) via a secure login mechanism.

1. Authentication & Authorization:

- a. **Admins & Agents:** Must log in using Phone Number and OTP . OTP is generated and sent via Twilio SMS G with time-based expiry for security
- b. **Customers:** Access to the tracking data is public but restricted to "Read-Only". Customers cannot modify route history or delete records. To book a consignment, customers must authenticate or provide verified contact details (OTP verification).

2. Data Transmission:

- a. All communication between the **Mobile Interface**, the **Web Portal**, and the **Database** will be encrypted using **HTTPS (SSL/TLS)** protocols. This ensures that sensitive data, such as customer addresses and agent locations, cannot be intercepted during transmission.

3. Database Security:

- a. The database will be protected behind a firewall to prevent direct external access. Only the EagleEye backend application will have permission to execute SQL queries.

- b. Regular backups of the database will be scheduled to prevent data loss in case of hardware failure.

4. Physical Security:

- a. The physical machine (server) hosting the application will be managed by the cloud provider (e.g., AWS/Azure) with industry-standard physical security measures. The **Mobile Devices** used by agents are personal devices, but the application will require re-authentication if the session expires to prevent unauthorized use if the device is lost or stolen.

4.0 Software Design

This section describes the architectural and detailed design of the *Eagle Eye Logistics: Web-Based Courier & Consignment Tracking System*.

4.1 System Architecture

The Eagle Eye Logistics System follows a **Hybrid Cloud-Based Three-Tier Architecture** with integrated external services.

>> **Client Layer:** This layer provides user interaction interfaces.

a) Web Portal

- Customers (Track Consignment, Book Parcel)
- Admins (Manage Hubs, Monitor Network, Send Broadcast Alerts)

b) Responsive Web View

- Delivery Agents (Update Status, Capture Signature, Scan Barcode)

>>**Application Layer:**

This is the core processing engine of the system.

Core Modules:

a) Logistics Core Engine

- Handles consignment lifecycle
- Generates AWB numbers
- Validates hub mappings
- Updates parcel status

b) Tracking & Route Module

- Fetches hub coordinates
- Interacts with Map API
- Renders parcel route timeline

c) Notification Service

- Sends automated Email/SMS alerts
- Triggers notifications on status change
- Manages broadcast messaging

d) Authentication & Authorization Module

- Implements Role-Based Access Control (RBAC)
- Manages login sessions using secure tokens
- Verifies user privileges before action execution

>>Data Layer:

This layer is responsible for persistent storage and retrieval of system data.

Database:

- Relational Database (MySQL / PostgreSQL)
- Stores:
 - User records
 - Hub data
 - Consignment records
 - Tracking history
 - Digital proof of delivery

Data Characteristics:

- Transactional integrity
- Indexed AWB lookup for fast search
- Foreign key constraints for relational consistency

>>Integration Layer:

a) Map Service API (OpenStreetMap / Leaflet)

- Renders tracking routes
- Displays hub locations
- Generates polyline path between hubs

b) SMS/Email Gateway

- Sends delivery confirmations
- Sends broadcast alerts
- Sends password reset messages

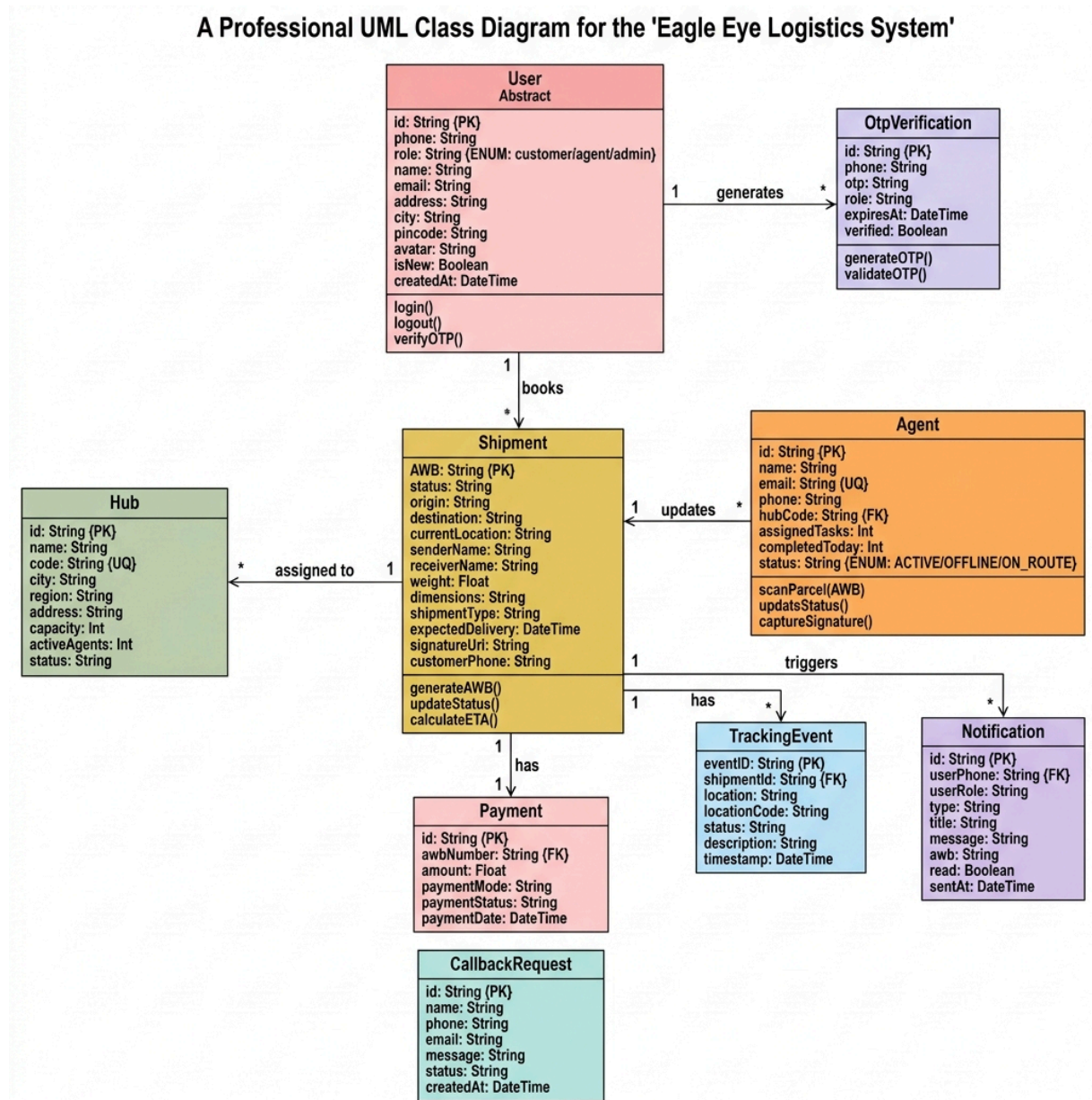
c) Cloud Hosting Environment

- Deployed on cloud infrastructure (AWS/Azure/Heroku)
- Supports auto-scaling and load balancing

4.2 Detailed Design

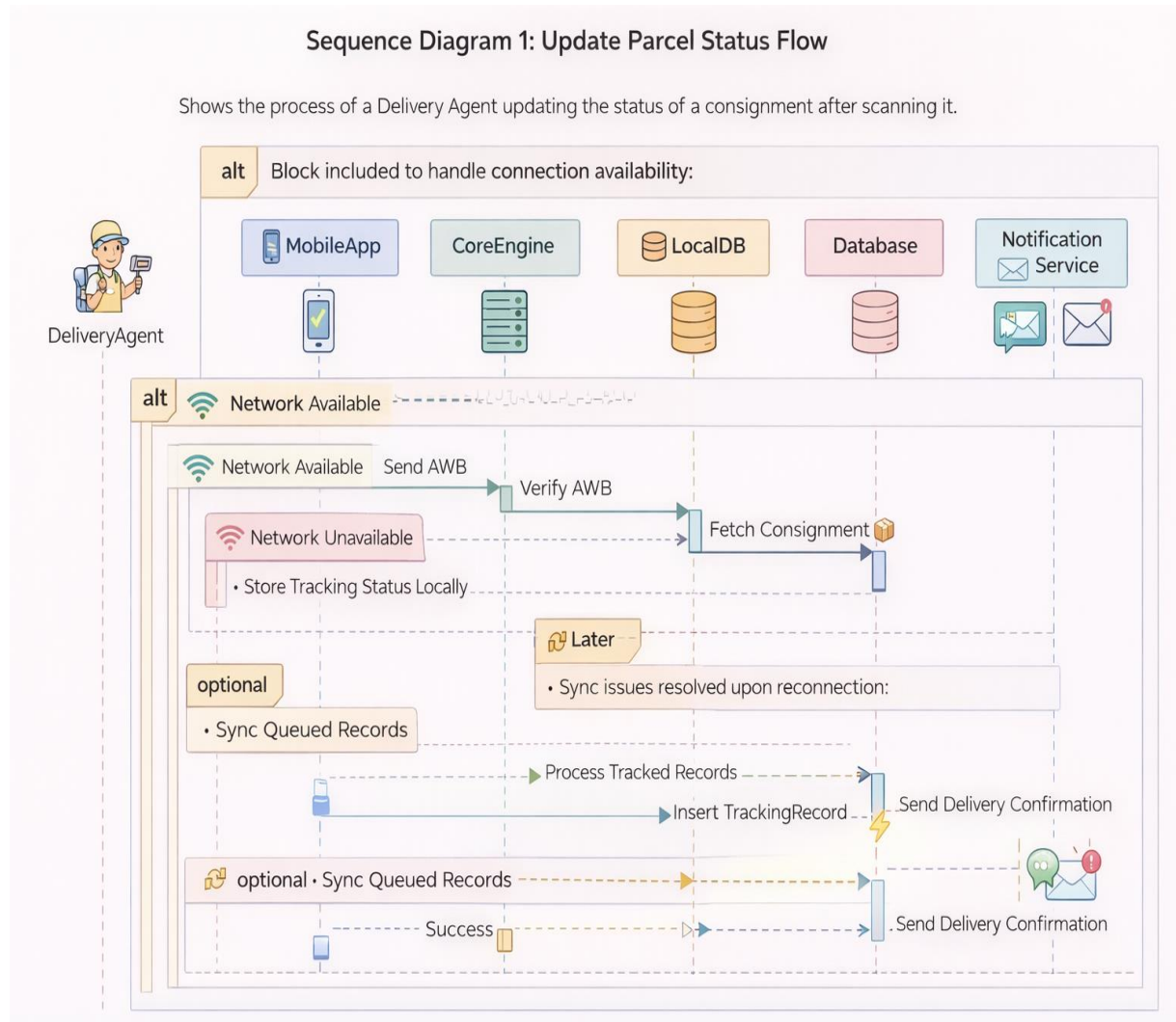
4.2.1 Class Diagram Design

The class diagram represents the static structure of the Eagle Eye Logistics System. It defines system entities, their attributes, methods, and relationships.



4.2.2 Sequence Diagrams

This diagram illustrates how a Delivery Agent updates parcel status and how the system ensures data consistency even during network failure.



4.3 Database Schema

The system uses a relational database to maintain data consistency, integrity, and traceability of logistics operations.

👤 1. Table: User

Column Name	Data Type	Key	Description
User ID	INT	PK	Unique user identifier
name	VARCHAR (100)		Full name of user
email	VARCHAR (100)	UQ	Unique email address
phone	VARCHAR (15)		Contact number
otpHash	VARCHAR (255)		OTP verification hash
role	ENUM		(Customer/Admin/Agent)

👤 2. Table: Customer

Column Name	Data Type	Key	Description
Customer ID	INT	PK	Unique customer ID
User ID	INT	FK	Reference to User(userID)
address	TEXT		Customer address
pincode	VARCHAR (10)		Postal code

Relationship:

Customer.userID → User.userID

👤 3. Table: Admin

Column Name	Data Type	Key	Description
adminID	INT	PK	Unique admin ID
userID	INT	FK	Reference to User(userID)

🚚 4. Table: DeliveryAgent

Column Name	Data Type	Key	Description
agentID	INT	PK	Unique agent ID
userID	INT	FK	Reference to User(userID)
assignedHub	INT	FK	Assigned Hub ID
availabilityStatus	ENUM		Active / Inactive

Relationship:

DeliveryAgent.userID → User.userID

DeliveryAgent.assignedHub → Hub.hubID

 5. Table: Hub

Column Name	Data Type	Key	Description
hubID	INT	PK	Unique hub ID
hubName	VARCHAR (100)		Hub name
address	TEXT		Hub location
pincode	VARCHAR (10)		Postal code
latitude	FLOAT		GPS latitude
longitude	FLOAT		GPS longitude

 6. Table: Consignment

Column Name	Data Type	Key	Description
AWB	VARCHAR (50)	PK	Unique Air Waybill Number
customerID	INT	FK	Reference to Customer
sourceHub	INT	FK	Source hub
destinationHub	INT	FK	Destination hub
weight	FLOAT		Parcel weight
currentStatus	VARCHAR (50)		Current tracking status
bookingDate	DATETIME		Booking timestamp
estimatedDelivery	DATETIME		Expected delivery time

Relationships:

Consignment.customerID → Customer.customerID

Consignment.sourceHub → Hub.hubID

Consignment.destinationHub → Hub.hubID

 7. Table: TrackingEvent

Column Name	Data Type	Key	Description
eventID	INT	PK	Unique tracking event ID
AWB	VARCHAR (50)	FK	Reference to Consignment
hubID	INT	FK	Location hub
agentID	INT	FK	Agent performing update

timestamp	DATETIME		Event time
activityType	VARCHAR (100)		Status description

Relationships:

TrackingEvent.AWB → Consignment.AWB

TrackingEvent.hubID → Hub.hubID

TrackingEvent.agentID → DeliveryAgent.agentID

8. Table: Notification

Column Name	Data Type	Key	Description
notificationID	INT	PK	Unique notification ID
userID	INT	FK	Receiver user
AWB	VARCHAR (50)	FK	Related consignment
message	TEXT		Notification message
type	ENUM		Email / SMS / Alert
sentAt	DATETIME		Sent timestamp

Relationships:

Notification.userID → User.userID

Notification.AWB → Consignment.AWB

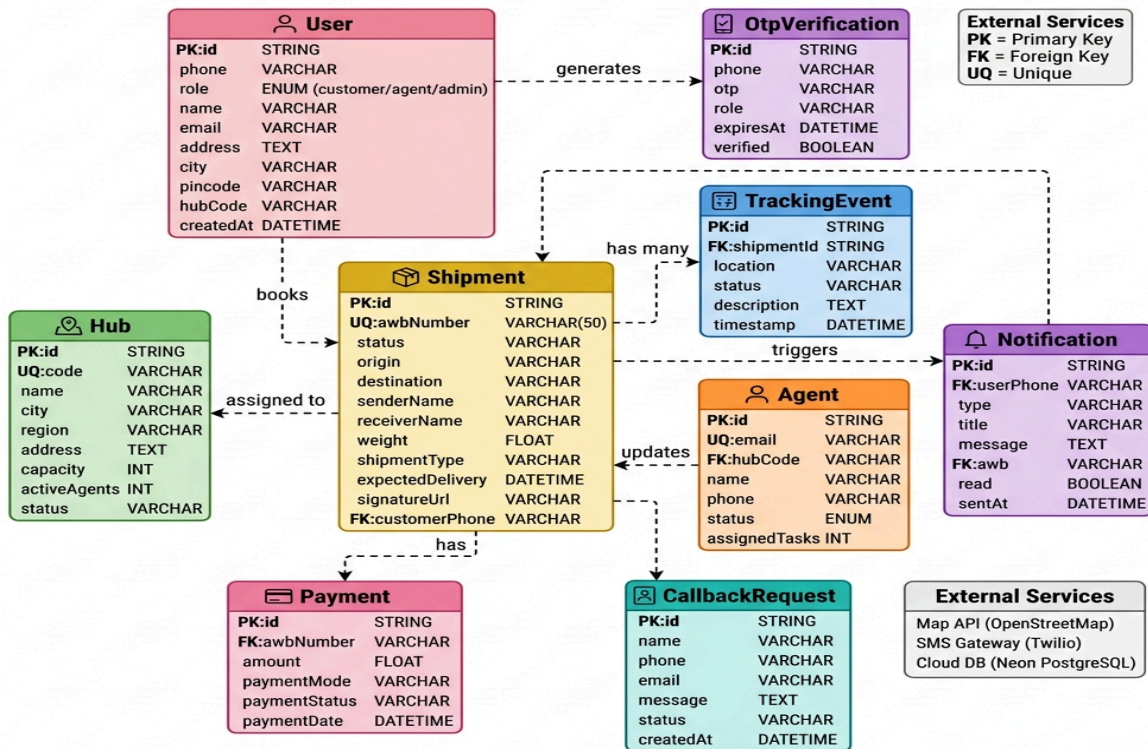
9. Table: Payment (Optional)

Column Name	Data Type	Key	Description
paymentID	INT	PK	Unique payment ID
AWB	VARCHAR (50)	FK	Related consignment
amount	FLOAT		Payment amount
paymentMode	VARCHAR (50)		UPI/Card/Cash
paymentStatus	VARCHAR (50)		Paid/Pending
paymentDate	DATETIME		Transaction date

Relationship:

Payment.AWB → Consignment.AWB

Database Schema - Eagle Eye Logistics System - ER Diagram



4.4 Security Design

4.4.1 Authentication and Authorization Mechanisms

1. User Authentication

The system implements secure login mechanisms to verify user identity.

Features:

- Phone Number + OTP based login
- OTPs verified using :
 - SHA-256 hashing
 - Salt-based encryption
- No plain-text password storage
- Account lock after multiple failed attempts

Optional Enhancement:

- OTP verification during:
 - Registration
 - Password reset

2. Role-Based Access Control (RBAC)

The system enforces strict role separation.

<i>Role</i>	<i>Permissions</i>
Customer	Book & Track Consignment (Read-only tracking)
Delivery Agent	Update parcel status, capture signature
Admin	Manage hubs, register agents, monitor system

Access control checks are performed:

- Before executing any update operation
- At API endpoint level

Unauthorized access attempts are logged.

3. Session Management

- LocalStorage-based session handling
- Token expiration time configured
- Session persistence via localStorage
- Automatic logout after inactivity

4.4.2 Security Protocols

1. Secure Communication

- All client-server communication via HTTPS
- SSL/TLS encryption enabled
- API endpoints protected via secure tokens

2. Database Security

- Foreign key constraints enforced
- Indexed AWB search for secure lookups
- Restricted database access (No public exposure)
- Database firewall enabled
- Regular backup & recovery mechanism

3. Protection Against Common Attacks

The system includes protection mechanisms against:

SQL Injection

- Parameterized queries
- ORM-based query handling

Cross-Site Scripting (XSS)

- Input sanitization
- Output encoding

Cross-Site Request Forgery (CSRF)

- CORS headers for API protection

Brute Force Attacks

- Rate limiting
- Login attempt threshold

4. Audit & Logging Mechanism

Every critical system action is logged:

- Consignment status update
- Admin configuration changes
- Agent activity logs
- Broadcast alerts

Audit logs include:

- User ID
- Timestamp
- IP address
- Action performed

This ensures accountability and traceability.

5. Notification Security

- Only authenticated users receive notifications
- Notification logs stored in database
- Sensitive information masked in SMS/email

5.0 Traceability

5.1 Requirement-to-Design Mapping

Requirement / Use Case ID	Requirement / Use Case Name	Design ID(s)	Design Artifact
UC-01	Book Consignment	D-CL-01, D-DB-01	Class Diagram, Database Schema
UC-02	Track Consignment	D-CL-01, D-SQ-02, D-DB-01	Class Diagram, Sequence Diagram
UC-03	Update Parcel Status	D-CL-01, D-SQ-01, D-DB-01	Class Diagram, Sequence Diagram
UC-04	Register Delivery Agent	D-CL-01, D-DB-01	Class Diagram, Database Schema
UC-05	Manage Hub	D-CL-01, D-ARCH-01	Class Diagram, Architecture Diagram
FR-01	Generate AWB	D-CL-01	Class Diagram
FR-02	Maintain Tracking History	D-DB-01	Database Schema
FR-03	Send Notification	D-CL-01, D-DB-01	Class Diagram, Database Schema
NFR-01	Secure Access	D-SEC-01	Security Design
NFR-02	Data Integrity	D-DB-01, D-SEC-02	Database & Security Design
NFR-03	System Scalability	D-ARCH-01	Architecture Design
NFR-04	Audit Logging	D-SEC-02	Security Design

>> Design ID Reference

- D-CL-01 → Class Diagram
- D-SQ-01 → Update Parcel Status Sequence Diagram
- D-SQ-02 → Track Consignment Sequence Diagram
- D-DB-01 → Database Schema (ER Diagram)
- D-SEC-01 → Authentication & Authorization
- D-SEC-02 → Security Protocols
- D-ARCH-01 → System Architecture

Github Link- <https://github.com/Anshk3671/EagleEye.git>